

Enhancing Insider Threat Detection Using Psychological Capital and Moral Foundations Theory

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I. PROBLEM STATEMENT

While the existing research on "Insider Threat Detection Using Natural Language Processing and Personality Profiles" provides a solid foundation [1], it does not fully capture the complexity of human behavior and decision-making in organizational contexts. The current approach lacks consideration of important psychological factors that could significantly improve the accuracy and comprehensiveness of insider threat detection. Our aim is to enhance the existing insider threat detection model by integrating two additional psychological frameworks: Psychological Capital (PsyCap) [2] and Moral Foundations Theory [3]. These frameworks will provide a more nuanced understanding of employee behavior, motivation, and decision-making processes, potentially leading to more accurate and earlier detection of insider threats.

II. APPROACH

Our approach will be integrating two additional psychological frameworks into the existing insider threat detection model:

- 1) *Psychological Capital (PsyCap)* [2]: We will incorporate the four key components of PsyCap - hope, efficacy, resilience, and optimism - into the detection system. This will allow us to assess the risk of employee disgruntlement, predict potential burnout, and monitor employee well-being over time.
- 2) *Moral Foundations Theory* [3]: We will analyze the five moral dimensions - Care/Harm, Fairness/Cheating, Loyalty/Betrayal, Authority/Subversion, and Sanctity/Degradation - to gain a better understanding of ethical decision-making processes, predict potential policy violations, and to assess organizational loyalty

A. Implementation

- 1) Develop linguistic markers for PsyCap [2] components and moral foundations dimensions [3]
- 2) Integrate these new markers into the existing natural language processing algorithm [1]. Our study would use the following datasets:
 - a) Carnegie Mellon: Insider Threat Test Dataset [4]
 - b) Enron Dataset [5]
- 3) Create a more comprehensive personality profile that includes PsyCap scores and moral foundations analysis.
- 4) Modify the existing decision tree to incorporate these new psychological factors.
- 5) If possible, develop a longitudinal tracking system to monitor changes in PsyCap and moral foundations over

time, which could indicate increased risk of insider threat behavior.

B. Evaluation

- Use the enhanced system to reanalyze the Enron dataset
- Compare the results with the original system to assess improvements in accuracy and detection rates

III. ANTICIPATED RESULTS

We anticipate several key outcomes from this enhanced insider threat detection system. Firstly, we expect to see improved accuracy in identifying insider threats, particularly for disgruntled employees and cases of insider collusion. The integration of Psychological Capital and Moral Foundations Theory should provide a more nuanced understanding of employee behavior, leading to earlier identification of potential insider threats.

Furthermore, we anticipate that the system will offer better differentiation between inadvertent insiders and those with malicious intent, thanks to the moral foundations analysis. The enhanced model is expected to provide a more comprehensive understanding of the psychological factors contributing to insider threats, which could have applications beyond threat detection. For instance, the insights gained could be valuable for improving overall employee well-being and organizational culture.

Lastly, we believe this approach represents a more ethical and comprehensive method for insider threat detection. By considering the complex psychological factors that influence human behavior in organizational settings, we can move beyond simplistic models and develop more sophisticated, context-aware threat detection systems. This holistic approach not only promises to enhance organizational security but also offers the potential to foster a more positive and supportive work environment.

REFERENCES

- [1] S. Eftimie, R. Moinescu, and C. Răcuciu, "Insider threat detection using natural language processing and personality profiles," in *2020 13th International Conference on Communications (COMM)*, 2020, pp. 325–330.
- [2] M. ÇAVUŞ and A. Kapusuz, "Psychological capital: Definition, components and effects," *British Journal of Education, Society Behavioural Science*, vol. 5, pp. 244–255, 03 2015.

- [3] J. Graham, J. Haidt, S. Koleva, M. Motyl, R. Iyer, S. P. Wojcik, and P. H. Ditto, "Chapter two - moral foundations theory: The pragmatic validity of moral pluralism," ser. *Advances in Experimental Social Psychology*, P. Devine and A. Plant, Eds. Academic Press, 2013, vol. 47, pp. 55–130. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/B9780124072367000024>
- [4] B. Lindauer, "Insider Threat Test Dataset," 9 2020. [Online]. Available: <https://kithub.cmu.edu/articles/dataset/InsiderThreatTestDataset/12841247>
- [5] B. Klimt and Y. Yang, "The enron corpus: A new dataset for email classification research," vol. 3201, 09 2004, pp. 217–226.